

Problem

Using the y parameters, derive formulas for Z_{in} , Z_{out} , A_i , and A_v for the common-emitter transistor circuit.

Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

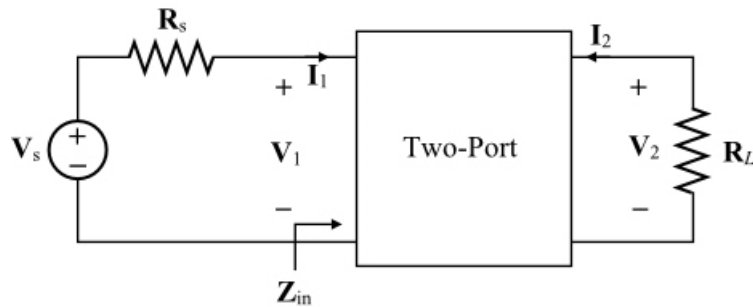


Figure 1

Step 2 of 8

Write the y -parameter equations.

$$I_1 = y_{11} V_1 + y_{12} V_2 \dots\dots (1)$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \dots\dots (2)$$

From the Figure 1, the current through load resistor:

$$I_2 = \frac{-V_2}{R_L}$$

Substitute $\frac{-V_2}{R_L}$ for I_2 in the equation (2).

$$\frac{-V_2}{R_L} = y_{21} V_1 + y_{22} V_2$$

$$-\left(\frac{1}{R_L} + y_{22}\right) V_2 = y_{21} V_1$$

$$V_2 = -\frac{y_{21} V_1}{\left(\frac{1}{R_L} + y_{22}\right)} \dots\dots (3)$$

Step 3 of 8

Substitute $-\frac{y_{21}V_1}{\left(\frac{1}{R_L}+y_{22}\right)}$ for V_2 in the equation (1).

$$\begin{aligned}
 I_1 &= y_{11}V_1 + y_{12}\left(-\frac{y_{21}V_1}{\left(\frac{1}{R_L}+y_{22}\right)}\right) \\
 &= \frac{\frac{y_{11}V_1}{R_L} + (y_{11}y_{22} - y_{12}y_{21})V_1}{\frac{1}{R_L} + y_{22}} \\
 &= \frac{y_{11}V_1Y_L + (y_{11}y_{22} - y_{12}y_{21})V_1}{Y_L + y_{22}} && \left(\text{since } Y_L = \frac{1}{R_L}\right) \\
 &= \left(\frac{y_{11}Y_L + \Delta_y}{Y_L + y_{22}}\right)V_1 && \left(\text{since } \Delta_y = y_{11}y_{22} - y_{12}y_{21}\right)
 \end{aligned}$$

Step 4 of 8

Write the expression for input impedance.

$$Z_{in} = \frac{V_1}{I_1}$$

Substitute $\left(\frac{y_{11}Y_L + \Delta_y}{Y_L + y_{22}}\right)V_1$ for I_1 in the above equation.

$$\begin{aligned}
 Z_{in} &= \frac{V_1}{\left(\frac{y_{11}Y_L + \Delta_y}{Y_L + y_{22}}\right)V_1} \\
 &= \frac{Y_L + y_{22}}{y_{11}Y_L + \Delta_y}
 \end{aligned}$$

Therefore, the input impedance, Z_{in} is $\boxed{\frac{Y_L + y_{22}}{y_{11}Y_L + \Delta_y}}$.

Step 5 of 8

Calculate the current gain expression.

$$\begin{aligned}
 A_i &= \frac{I_2}{I_1} \\
 &= \frac{y_{21}V_1 + y_{22}V_2}{I_1} \\
 &= y_{21}Z_{in} + \frac{y_{22}}{I_1} \left(-\frac{y_{21}V_1}{\left(\frac{1}{R_L} + y_{22}\right)} \right) \\
 &= y_{21}Z_{in} - \frac{y_{22}y_{21}Z_{in}}{Y_L + y_{22}}
 \end{aligned}$$

Substitute $\frac{Y_L + y_{22}}{y_{11}Y_L + \Delta_y}$ for Z_{in} in the above equation.

$$\begin{aligned}
 A_i &= \left(y_{21} - \frac{y_{22}y_{21}}{Y_L + y_{22}} \right) \left(\frac{Y_L + y_{22}}{y_{11}Y_L + \Delta_y} \right) \\
 &= \left(\frac{y_{21}(Y_L + y_{22}) - y_{22}y_{21}}{Y_L + y_{22}} \right) \left(\frac{Y_L + y_{22}}{y_{11}Y_L + \Delta_y} \right) \\
 &= \frac{y_{21}Y_L + y_{21}y_{22} - y_{22}y_{21}}{y_{11}Y_L + \Delta_y} \\
 &= \frac{y_{21}Y_L}{y_{11}Y_L + \Delta_y}
 \end{aligned}$$

Therefore, the voltage gain expression, A_i is $\boxed{\frac{y_{21}Y_L}{y_{11}Y_L + \Delta_y}}$.

Step 6 of 8

Calculate the voltage gain expression.

$$A_v = \frac{V_2}{V_1}$$

Substitute $-\frac{y_{21}V_1}{\left(\frac{1}{R_L} + y_{22}\right)}$ for V_2 in the above equation.

$$\begin{aligned}
 A_v &= \frac{-\frac{y_{21}V_1}{\left(\frac{1}{R_L} + y_{22}\right)}}{V_1} \\
 &= -\frac{y_{21}V_1}{\frac{1}{R_L} + y_{22}} \\
 &= -\frac{y_{21}V_1}{Y_L + y_{22}}
 \end{aligned}$$

Therefore, the voltage gain expression, A_v is $\boxed{-\frac{y_{21}V_1}{Y_L + y_{22}}}$.

Step 7 of 8

To calculate output impedance, short the input source and connect a test voltage source across the load. Draw the modified circuit.

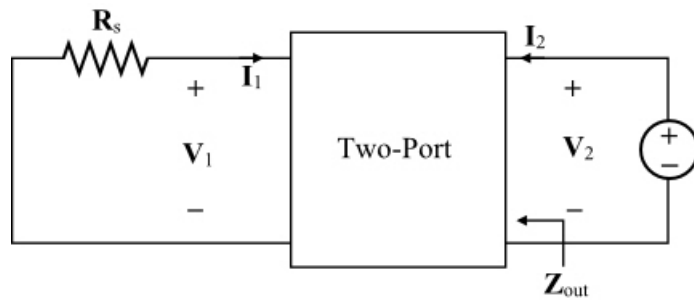


Figure 2

Write the output impedance expression.

$$\begin{aligned} Z_{\text{out}} &= \frac{V_2}{I_2} \\ &= \frac{V_2}{Y_{21}V_1 + Y_{22}V_2} \\ &= \frac{1}{Y_{21} \frac{V_1}{V_2} + Y_{22}} \dots\dots (4) \end{aligned}$$

Apply Kirchhoff's voltage law to left side loop.

$$R_s I_1 + V_1 = 0$$

$$V_1 = -R_s I_1$$

Substitute $-R_s I_1$ for V_1 in the equation (1).

$$I_1 = y_{11}(-R_s I_1) + y_{12}V_2$$

$$I_1 + R_s y_{11} I_1 = y_{12}V_2$$

$$I_1 = \frac{y_{12}V_2}{1 + R_s y_{11}}$$

$$-\frac{V_1}{R_s} = \frac{y_{12}V_2}{1 + R_s y_{11}}$$

$$\frac{V_1}{V_2} = -\frac{y_{12}R_s}{1 + R_s y_{11}}$$

Substitute $-\frac{y_{12}R_s}{1 + R_s y_{11}}$ for $\frac{V_1}{V_2}$ in the equation (4).

$$\begin{aligned} Z_{\text{out}} &= \frac{1}{Y_{21} \left(-\frac{y_{12}R_s}{1 + R_s y_{11}} \right) + Y_{22}} \\ &= \frac{1 + R_s y_{11}}{Y_{22} + R_s y_{11} Y_{22} - y_{12} Y_{21} R_s} \\ &= \frac{1 + R_s y_{11}}{Y_{22} + \Delta_y R_s} \\ &= \frac{\frac{1}{R_s} + y_{11}}{\frac{Y_{22}}{R_s} + \Delta_y} \\ &= \frac{Y_s + y_{11}}{Y_{22} Y_s + \Delta_y} \end{aligned}$$

Therefore, the output impedance of the common emitter transistor circuit, Z_{out} is

$$\boxed{\frac{Y_s + y_{11}}{Y_{22} Y_s + \Delta_y}}.$$